



PURE PERFORMANCE BATTLE

Mithun Mohandas | mithun@digit.in

W

hat's the most important part of your PC? The processor, without a doubt. And

we are, right now, in a golden period where manufacturers are leapfrogging over one another to bring out the best that they can. So if you're wondering which is the best for you, then read on.

UNDERLYING TECH

Both Intel and AMD have been at the forefront of processor technology for donkey's years. Intel created the x86 architecture back in 1978 which is what the entire PC industry has been using for the longest time and AMD came out with the 64-bit version in 1999 (implemented in 2003) which then was adopted by everyone else and is currently the dominant instruction set. Both companies have had processor families that have been the best of that generation. Intel has had the larger piece of the pie for a longer time since the beginning. Their performance leadership was

disrupted when AMD introduced the K7 architecture in the form of the Athlon processor family back in 1999.

Intel would soon snatch back the leadership within three years. And it was only in 2017 that AMD resurfaced with their new Ryzen processors and knocked it out of the park with a ridiculous amount of cores. Single-core performance was still behind Intel but multi-core performance was miles ahead of Intel. This was followed by a few years of back-and-forth with Intel or AMD leading the performance leadership. Then Intel unveiled their hybrid architecture to much fanfare and has been optimising it since then. As it stands now, the two companies have extremely competitive flagships which are doing wonders for the average PC buyer. Everyone is eyeing 2025 because that's when Intel's CEO believes that his company will attain process performance leadership. That's because Intel has been aggressively upping their process technology

PRODUCTIVITY / OFFICE WORK

As it stands now, both AMD and Intel have great processors for simple productivity needs. Both have a diverse stack of processors starting from processors that cost about 5K. There's no need to buy flagship processors if you're simply going to be browsing the web, sending emails and sifting through spreadsheets. If you're looking to use the most recent technologies from both companies, then those are usually relegated to the mid-tier processors. The low-end processors usually don't feature the good stuff. Nevertheless, your needs will be handled regardless of which brand you decide to go with.

GAMING

Gaming is where AMD has the advantage. AMD's implementation of a massive L3 cache thanks to silicon stacking which they market as X3D gives them a significant advantage with gaming since games favour large caches.